

I Have Seen The Future, And I Am Scared

Although the threat was characterised as relatively low, anti-virus companies said on August 2, 2004, that a Trojan program for Pocket PCs has emerged. The program, known as Backdoor.Brador.A or WinCE.Brador.a, lets an attacker gain full control of the infected handheld device. Brador is the first 'backdoor Trojan' program that has emerged for Pocket PCs, thus far. Such programs cannot self propagate, and have to be sent as e-mail attachments. Symantec said Brador seems to be limited to devices that use ARM processors and Pocket PC operating systems.

Eugene Kaspersky, head of anti-virus research at Kaspersky Labs, said in a statement, "We can be sure that the computer underground will snatch at the chance to attack PDAs and mobile phones in the nearest future."

virus threat to cell phone networks was "probably" five times larger than that to PCs, because of how few people are focused on the problem. The article quoted incidents in Japan, when DoCoMo customers "were being sent messages that froze their screens and automatically dialled 110—the hotline to the police in Japan." If they could dial the police, they could just dial other numbers as well, at random, racking up a huge phone bill?

As smartphones replace laptops, the problem will intensify. Right now, infection may mean just be the loss of your contacts and a few notes; later, it could crash your PC.

Wireless Networks

The issue is that people are at risk without knowing it. It's a time bomb. In a wide-range WiMax wireless network, where people typically sit at a large distances from access points, they may be at risk even when they are not connected to the Internet—the network is there for anyone to exploit. There's a spectrum of threats: PANs—typically Bluetooth-connected areas; WLANs; and WANs—WiMax networks. Few people are aware of the risks of insecure networks.

Smartphones, as also other Bluetooth-enabled devices, could be used as a backdoor into corporate networks: when an infected phone is synchronised with a terminal, the entire office network could get infected.

According to research agency Gartner, WLAN attacks will be the biggest wireless-related security issues through to 2008. The current insecurity of WLANs is well known and well documented—not too many administrators take the trouble of securing them; several independent researchers have gone to the 40th floor, and easily accessed a WLAN on the 38th floor, as benign experiments.

Wireless Internet Magazine warned in 2001 of "economic information warfare, involving sophisticated attacks against entire economies, commerce and enterprises"; and "next-generation computer viruses that will self-mutate and are designed to destabilize, confuse and destroy critical electronic infrastructure essential to industry and government." These are plausible scenarios, and WLAN security, or the lack of it, will hurt us.

VoIP

Anything that's smart can be subverted. You are soon going to have smart VoIP—something that will trickle down into every device. This is a perfect recipe for disaster.

We can be sure that the computer underground will snatch at the chance to attack PDAs and mobile phones in the nearest future.

Eugene Kaspersky
Head of Anti-virus
Research
Kaspersky Labs

It is envisaged that much of the role the existing telephone system plays will be replaced by VoIP networks. Imagine your home phone line replaced with a fixed VoIP line, operated by your broadband provider. Data and voice will be on the same cable. Your PC will, naturally, be connected to this cable. What happens if your computer gets infected? It will suck up all the bandwidth, and you can't even place a call to customer service! And if the attack happens at the service provider's end? You will end up with no phone, no Internet service, nothing.

The problem is that VoIP networks happen to be especially easy to crack. It's hard to secure a simple protocol like http, so what are the chances of securing the whole range of protocols used in VoIP, and written by different groups? That's what VoIP is—three complex protocols, each of them not a single protocol, but a protocol family in its own right.

Consider Asterix, a PBX software for Linux. PBX is short for Private Branch eXchange, and is a small, private telephone exchange. With just a configurable tool such as Asterix in a hacker's hands—not to mention various, other underground tools—imagine the damage that could be inflicted on a VoIP network.

VoIP, where you can send any return number you want, takes phishing to a new level. (Phishing is the sending of spam that makes you believe you're receiving it from a legitimate source, and which asks you to disclose personal financial details such as your bank account number.) You could be phished by a person with a PDA and a headset sitting outside your home.

Cable modems run real-time OSes; they can be hacked and reprogrammed. Eventually, VoIP phones will make use of a cable-modem type device. With such a device sitting in your home, it will definitely get exciting for hackers.

At some stage, you could imagine your 'VoIP line' connected to your desktop, laptop, TV, entertainment system, and more. Someone could send you a pre-recorded message over the channel, and cause your phone to crash. And every device that's connected to the channel is suddenly under threat.

Consider the convergence of devices in the home that's happening. You are soon going to have a single device that acts as entertainment centre, VCR, TV, PC, and more. This device is connected to a broadband cable, and suddenly, is open to hacking.

Convergence also means Bluetooth-enabled refrigerators, washing machines, microwave ovens. All connected via the cable, all vulnerable, and all putting the others at risk. Soon, the term 'virus' will take on a new meaning, not limited to 'computers': any device with a microchip and software will be at risk.

As we take small, hesitant steps into the future, let us hope we can be smart enough to prevent rather than cure. ☒

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